The Geometric Signature of Primes in Farey Sequences

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Abstract

We discover that each prime number leaves a measurable geometric signature in the distribution of fractions. Decomposing the uniformity of the Farey sequence one integer at a time, we find that primes and composites play fundamentally different roles: composites account for ∼96% of all uniformity improvement while primes are responsible for ∼99% of all uniformity *damage*—and the direction of each prime's geometric effect is controlled by the Mertens function $M(p) = \sum_{k \le p} \mu(k)$, a global arithmetic quantity encoding the factorization history of all integers up to p. This creates a direct, previously unknown bridge between the geometry of rational numbers and multiplicative number theory.

We prove this connection through exact identities, including a universal formula evaluating the Farey exponential sum at every integer frequency in terms of the Mertens function at divisor-scaled arguments (Theorem 3.5), and an Injection Principle showing each prime inserts its fractions into distinct Farey gaps via modular inversion (Theorem 5.2). The identities reveal a striking compression phenomenon: the bridge identity collapses $\sim 3p^2/\pi^2$ geometric data points into a single integer $M(p)$, with the sign of M predicting the geometric outcome at 99.98% accuracy when $M(p) < 0$. A four-term algebraic decomposition shows that the new-fraction discrepancy nearly cancels the dilution benefit ($\mathcal{D}/A \to 1$), leaving the Mertens-controlled cross term to tip the balance.

All core identities (139 results across nine Lean 4 files, zero `sorry`) are formally verified, with the bridge-identity proof chain constructed autonomously by the Aristotle theorem prover. Among 9,588 primes $p \ge 11$ up to 100,000, only one ($p = 92{,}173$, certified by 256-bit interval arithmetic) has $\Delta W(p) > 0$ despite $M(p) < 0$; for the $M(p) \le -3$ class (4,617 primes), there are zero counterexamples.

Keywords: Farey sequence, Mertens function, rank deviation, Ramanujan sums, formal verification, Lean 4.
MSC 2020: 11B57, 11N37, 11A25, 11Y35.

1 Introduction

1.1 Farey sequences and their discrepancy

For a positive integer N, the *Farey sequence* $\mathcal{F}_N$ is the ascending list of fractions a/b with $0 \le a \le b \le N$ and $\gcd(a, b) = 1$. Its cardinality is

$$|\mathcal{F}_N| = 1 + \sum_{k=1}^{N} \varphi(k). \tag{1}$$

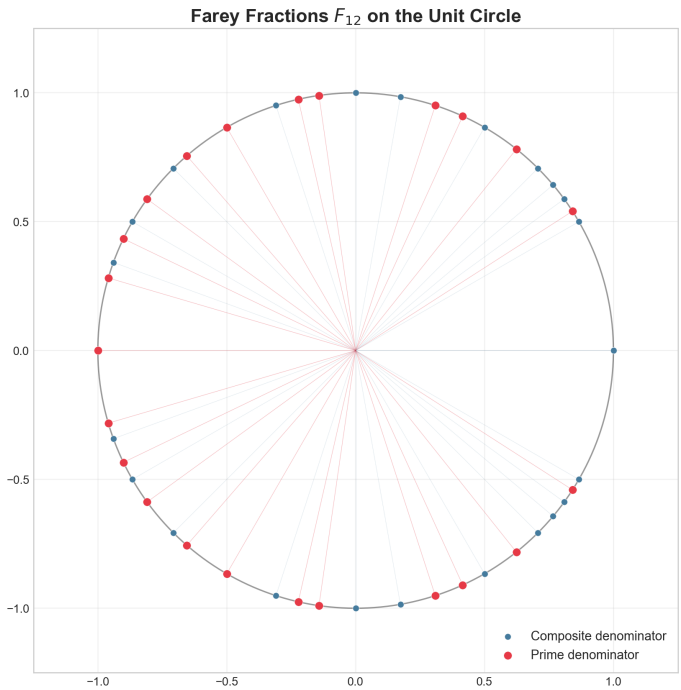

Figure 1: The Farey sequence $\mathcal{F}_{12}$ on the unit circle: each fraction a/b is placed at angle $2\pi a/b$. The pattern is strikingly uniform, with the largest gaps near 0 and 1 (top of the circle) where the denominators are smallest.

Writing $n = |\mathcal{F}_N|$ and listing elements as $f_0 < f_1 < \cdots < f_{n-1}$, the *squared rank deviation* (which we call the *wobble*, following the Franel tradition) is

$$W(N) \;=\; \sum_{j=0}^{n-1} \left(f_j - \frac{j}{n} \right)^2 . \tag{2}$$

Remark 1.1 (Interval vs. circle)**.** The Farey sequence lives on the *closed interval* $[0, 1]$, with $f = 0$ and $f = 1$ as distinct elements. Throughout this paper we use circle diagrams (identifying $[0, 1)$ with $\mathbb{R}/\mathbb{Z}$) as a visualization aid, but the discrepancy (2) and all identities in Section 3 are stated and proved on the interval, where the two boundary points $f = 0$ and $f = 1$ contribute separately. In particular, the $+2$ in the bridge identity (Theorem 3.1) arises from these two distinct boundary contributions.

1.2 The Franel–Landau theorem

Franel [3] and Landau [6] proved that the Riemann Hypothesis is equivalent to the bound $\sum_{j=0}^{n-1} |f_j - j/n| = O(N^{1/2+\varepsilon})$ for every $\varepsilon > 0$, where $n = |\mathcal{F}_N|$. Equivalently (see Hardy–Wright [10], Ch. XVIII), RH holds if and only if $W(N) = O(N^{-1+\varepsilon})$. Thus the quantity $W(N)$ defined above—which we call the *wobble*, following the Franel-type squared rank deviation—encodes information about the non-trivial zeros of the Riemann zeta function.

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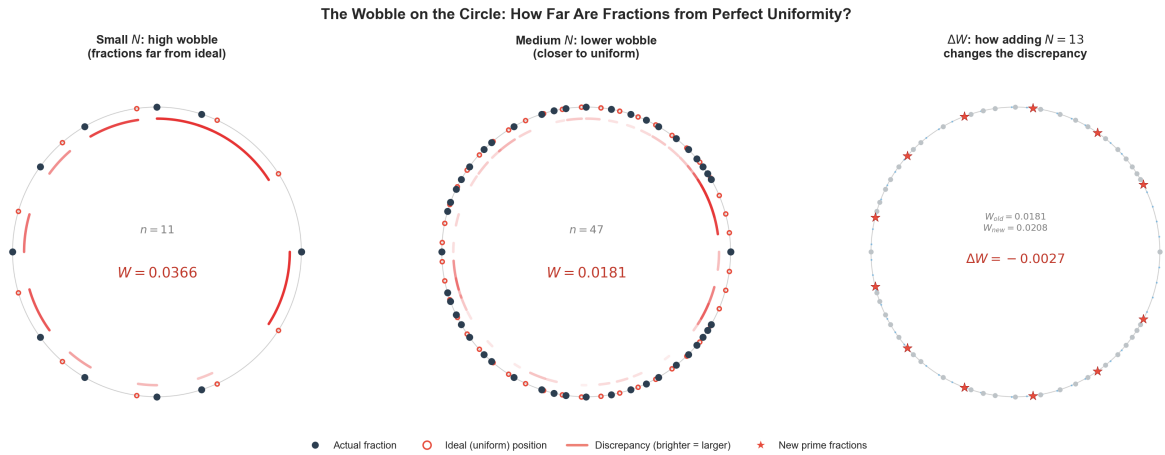

Figure 2: The wobble visualized on the circle. Dark dots show actual Farey fractions; red circles show ideal (equally spaced) positions. Red arcs connect each fraction to its ideal position—brighter arcs mean larger discrepancy. **Left:** $\mathcal{F}_5$ has high wobble (large gaps between actual and ideal). **Center:** $\mathcal{F}_{12}$ has lower wobble (fractions closer to uniform). **Right:** adding prime $p = 13$ (red stars) changes the ideal positions and the wobble.

1.3 The observation that started it all

Place the Farey fractions of $\mathcal{F}_N$ as points on the unit circle at angles $2\pi f$. As N grows, new fractions appear and the circle fills in. The first author observed a striking asymmetry: when a *prime* p is added, every one of the $p-1$ new fractions k/p lands on the circle at perfectly equispaced positions, filling voids uniformly across the entire circle. This happens because every integer below a prime is coprime to it ($\varphi(p) = p - 1$), so no position is skipped. Composites, by contrast, skip positions whose numerators share a factor with the denominator, leaving voids in some regions while clustering in others.

Moreover, by the standard mediant property of Farey sequences, each new fraction k/p is the *mediant* of its two neighbors in $\mathcal{F}_{p-1}$—the geometrically optimal insertion point. This means primes do not merely add points; they add them in the most balanced way possible.

These geometric observations led to a natural question: does the uniformity of the Farey sequence on the circle improve or worsen at each step, and what controls the answer?

1.4 Per-step decomposition: a new perspective

Despite decades of work on the Franel–Landau framework, the *per-step* behavior—how W changes when a single integer is appended—appears never to have been studied. We define

$$\Delta W(N) \;=\; W(N-1) - W(N), \tag{3}$$

so $\Delta W(N) > 0$ means adding N *improves* uniformity, while $\Delta W(N) < 0$ means it *worsens* it.

The per-step view reveals a sharp asymmetry invisible in the cumulative $W(N)$: composites almost always improve uniformity (96% of composites $N \leq 200$ have $\Delta W > 0$), while primes are the primary source of "damage" (99% of the total negative ΔW mass comes from primes, in computations through $N = 200{,}000$). Moreover, for primes, the sign of $\Delta W(p)$ tracks the Mertens function $M(p) = \sum_{k \leq p} \mu(k)$—the Mertens function acts as a "dial" controlling the prime contributions. This creates a direct bridge between two worlds that had no known connection

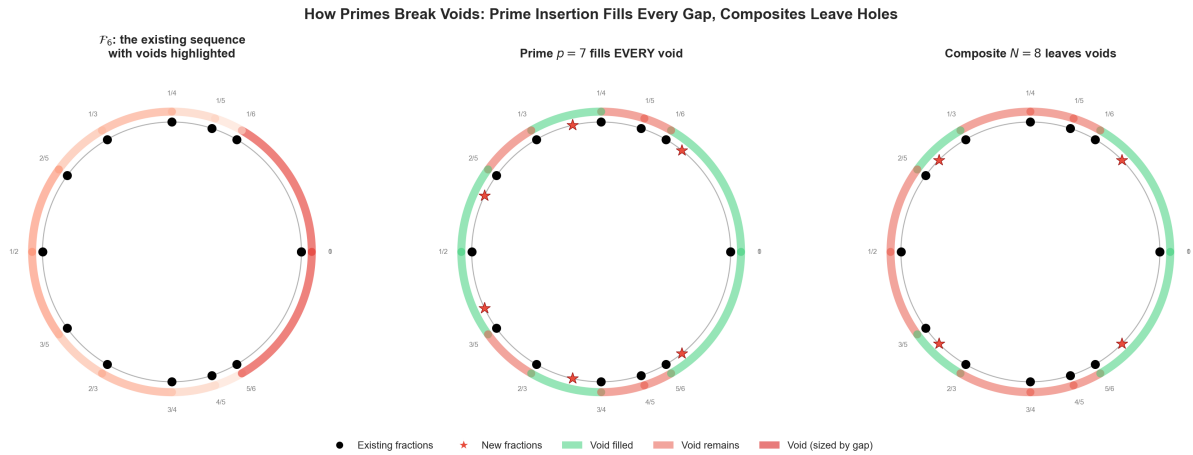

Figure 3: How primes break voids. **Left:** $\mathcal{F}_6$ with voids highlighted by size (darker = larger). **Center:** prime $p = 7$ inserts one fraction into each of $p - 1 = 6$ distinct gaps (green arcs), filling some voids and skipping others. **Right:** composite $N = 8$ leaves voids unfilled (red arcs) due to missing coprime numerators.

at this level of granularity: the geometry of Farey discrepancy and the arithmetic of the Möbius function.

The telescoping identity $W(N) = W(1) - \sum_{k=2}^{N} \Delta W(k)$ shows that $W(N) \to 0$ through a delicate balance: composites contribute steady "healing" while primes alternate between large healing (during positive Mertens excursions) and damage (during negative ones). This decomposition of Farey convergence into prime and composite contributions was previously unknown.

2 Definitions and Notation

Definition 2.1 (Rank discrepancy)**.** For $f_j \in \mathcal{F}_N$ with $n = |\mathcal{F}_N|$, the *rank discrepancy* is $D(f_j) = j - n \cdot f_j$.

Definition 2.2 (Shift)**.** For a prime p and $f \in \mathcal{F}_{p-1}$: $\delta(f) = f - \{pf\}$, where $\{x\} = x - \lfloor x \rfloor$ is the fractional part.

Definition 2.3 (Mertens function)**.** $M(N) = \sum_{k=1}^{N} \mu(k)$, where μ is the Möbius function.

3 New Identities

The identities in this section follow from classical Ramanujan-sum decompositions [9, 12] combined with the Farey involution and the Möbius function. The evaluation $\sum_{f \in \mathcal{F}_N} e^{2\pi i f} = M(N) + 1$ is implicit in the Farey-series chapter of Edwards [8] (see also Hardy–Wright [10], Ch. XVI). The per-frequency generalization (Theorems 3.3 and 3.5) and the per-step application to $\Delta W(N)$ (Theorems 3.9–3.14) appear, to our knowledge, not to have been stated previously in this form. All identities and supporting lemmas are formally verified in Lean 4 (139 results across nine files, zero `sorry`); all prime-parameter identities are also verified computationally for all primes $p \leq 100{,}000$, and the universal formula (Theorem 3.5) is verified on the grid $m \leq 40$, $N \leq 30$.

3.1 The Bridge Identity

Theorem 3.1 (Bridge Identity)**.** *For every prime $p \geq 2$,*

$$\sum_{f \in \mathcal{F}_{p-1}} e^{2\pi i p f} \;=\; M(p) + 2. \tag{4}$$

Proof sketch. Decompose by denominator. The boundary terms $f = 0$ and $f = 1$ contribute $1 + 1 = 2$. For each denominator $2 \leq b \leq p - 1$, the inner sum over numerators coprime to b is the Ramanujan sum $c_b(p)$. Since $\gcd(p, b) = 1$ for $b < p$, we have $c_b(p) = \mu(b)$. Summing: $2 + \sum_{b=2}^{p-1} \mu(b) = 2 + M(p-1) - 1 = M(p) + 2$, using $M(p-1) = M(p) + 1$. □

Remark 3.2. Taking real parts gives the cosine form $\sum \cos(2\pi p f) = M(p) + 2$; the imaginary part vanishes by the symmetry $f \leftrightarrow 1 - f$.

The bridge identity is a special case of a more general result:

Theorem 3.3 (Generalized Bridge Identity)**.** *For any positive integer N and any prime $m > N$,*

$$\sum_{f \in \mathcal{F}_N} e^{2\pi i m f} \;=\; M(N) + 1. \tag{5}$$

In particular, all primes $m > N$ yield the same *exponential sum over $\mathcal{F}_N$, depending only on $M(N)$.*

Proof sketch. Same decomposition as Theorem 3.1. Every denominator $b \leq N < m$ satisfies $\gcd(b, m) = 1$, so $c_b(m) = \mu(b)$. The boundary gives 2; the interior gives $M(N) - 1$. Total: $M(N) + 1$. □

Remark 3.4. Theorem 3.1 is the case $N = p - 1$: $M(p-1) + 1 = (M(p) + 1) + 1 = M(p) + 2$. For tensor-product Farey sets in d dimensions, the sum at a prime-vector frequency is $(M(N) + 1)^d$; when $M(N) = -1$ (e.g., $N = 10$), this vanishes in every dimension.

Both Theorems 3.1 and 3.3 are special cases of a universal formula:

Theorem 3.5 (Universal Farey Exponential Sum)**.** *For any positive integers m and N,*

$$\sum_{f \in \mathcal{F}_N} e^{2\pi i m f} \;=\; M(N) + 1 + \sum_{\substack{d \mid m \\ d \geq 2}} d \cdot M(\lfloor N/d \rfloor)\,. \tag{6}$$

Proof sketch. Decompose by denominator as before: the sum equals $2 + \sum_{b=2}^{N} c_b(m)$. The Ramanujan sum satisfies $c_b(m) = \sum_{d \mid \gcd(b,m)} d\,\mu(b/d)$. Exchange the order of summation over b and d: for each divisor $d \mid m$ with $d \geq 2$, the contribution is $d \sum_{k=1}^{\lfloor N/d \rfloor} \mu(k) = d \cdot M(\lfloor N/d \rfloor)$. The $d = 1$ terms give $M(N) - 1$; adding the boundary 2 yields $M(N) + 1$. □

Remark 3.6. This is a *complete* evaluation of the Farey exponential sum at every integer frequency in terms of the Mertens function at divisor-scaled arguments. For prime $m > N$, the only divisor $d \geq 2$ is m itself, and $\lfloor N/m \rfloor = 0$ gives $M(0) = 0$, recovering Theorem 3.3. Verified computationally for all $m \leq 40$ and $N \leq 30$.

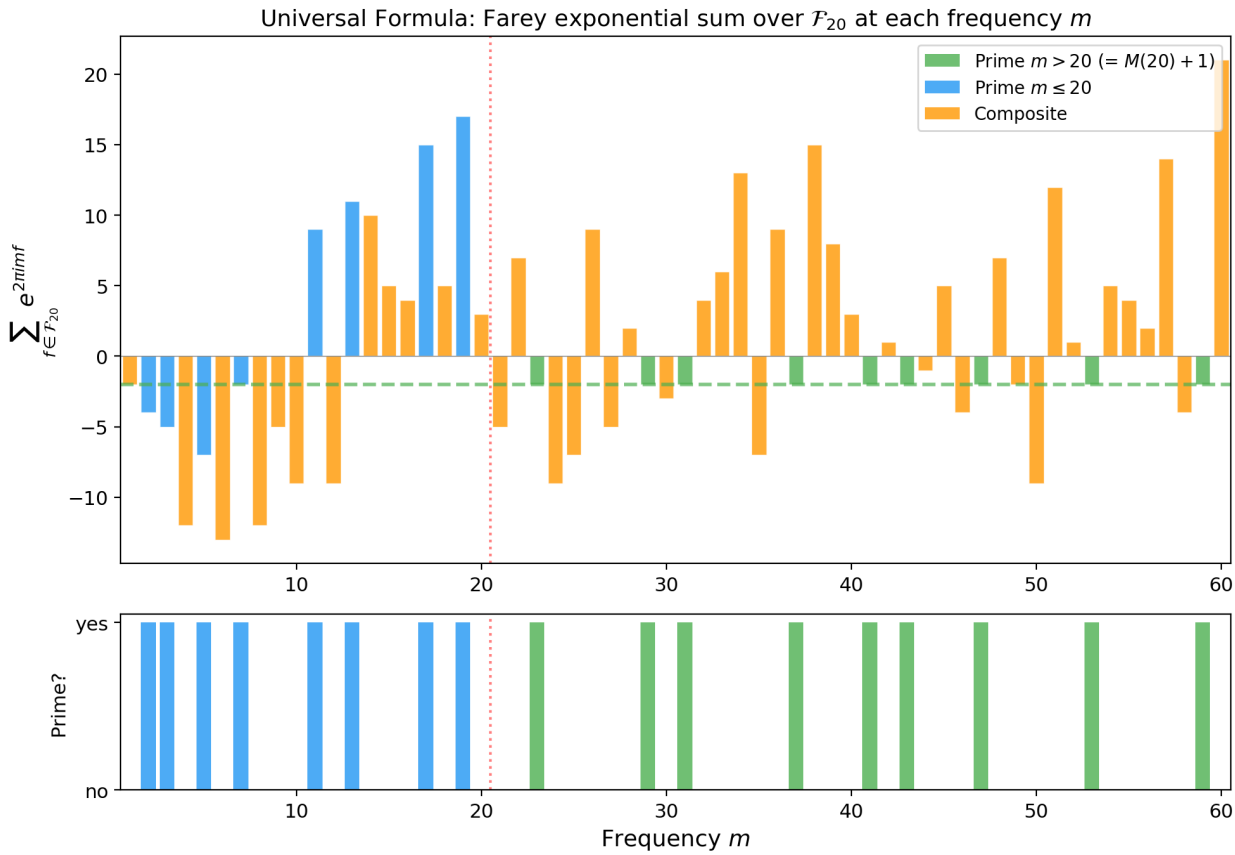

Figure 4: The universal formula for $\mathcal{F}_{20}$. **Top:** The Farey exponential sum at each frequency m. Green bars (primes $m > 20$) all equal $M(20) + 1 = -2$—they are *identical*, independent of which prime. Blue bars (primes $m \leq 20$) and orange bars (composites) vary. **Bottom:** prime indicator. The dashed line at $m = N = 20$ separates the two regimes.

3.2 Character-Weighted Bridge and the Generalized Riemann Hypothesis

The bridge identity extends naturally to Dirichlet characters.

Theorem 3.7 (Character-Weighted Bridge)**.** *For any Dirichlet character χ and any prime $m > N$,*

$$\sum_{f \in \mathcal{F}_N} \chi(\mathrm{denom}(f))\, e^{2\pi i m f} \;=\; 1 + \sum_{b=1}^{N} (\chi \cdot \mu)(b). \tag{7}$$

Proof sketch. Same denominator decomposition as Theorem 3.3. Each denominator b contributes $\chi(b) \cdot c_b(m) = \chi(b)\,\mu(b)$ when $\gcd(m, b) = 1$ (which holds for all $b \leq N < m$). Boundary: $\chi(1)(1 + 1) = 2$; interior: $\sum_{b=2}^{N} \chi(b)\mu(b)$. Total: $2 + \sum_{b=2}^{N} (\chi\mu)(b) = 1 + \sum_{b=1}^{N} (\chi\mu)(b)$. □

Remark 3.8. The Dirichlet series $\sum (\chi\mu)(n)\, n^{-s} = 1/L(s, \chi)$. Since the Generalized Riemann Hypothesis (GRH) for $L(s, \chi)$ is equivalent to $\sum_{b \leq N} (\chi\mu)(b) = O(N^{1/2+\varepsilon})$, the character-weighted bridge provides a Farey-geometric restatement of GRH for *every* Dirichlet L-function:

$$\text{GRH for } L(s, \chi) \iff \sum_{f \in \mathcal{F}_N} \chi(\mathrm{denom}(f))\, e^{2\pi i m f} = O(N^{1/2+\varepsilon}) \quad \text{for every prime } m > N.$$

The trivial character $\chi = 1$ recovers the RH characterization of Section 8.

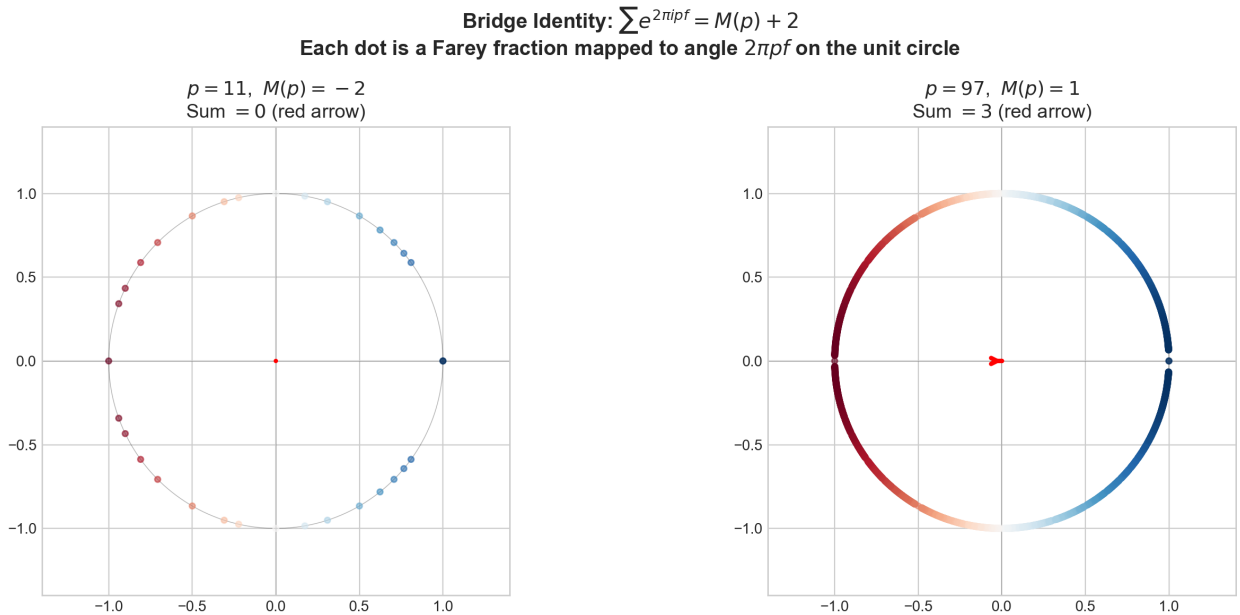

Figure 5: The Bridge Identity visualized. Each Farey fraction $f \in \mathcal{F}_{p-1}$ becomes a point at angle $2\pi p f$ on the unit circle. Their vector sum (red arrow) equals $M(p) + 2$. **Left:** $p = 11$, $M = -2$, sum $= 0$ (vectors cancel). **Right:** $p = 97$, $M = 1$, sum $= 3$ (net rightward bias).

3.3 The Master Involution Principle

Theorem 3.9 (Master Involution Principle)**.** *The Farey involution $\sigma(f) = 1 - f$ satisfies $D(\sigma(f)) = -D(f) - 1$. For any symmetric function g (i.e., $g(f) = g(1-f)$):*

$$\sum_{f \in \mathcal{F}_N} D(f)\, g(f) \;=\; -\frac{1}{2} \sum_{f \in \mathcal{F}_N} g(f). \tag{8}$$

Proof sketch. Substitute $f \mapsto \sigma(f)$ and use $g(\sigma(f)) = g(f)$, $D(\sigma(f)) = -D(f) - 1$. Adding the original and substituted sums gives $2 \sum Dg = -\sum g$. □

3.4 The Displacement–Cosine Identity

Theorem 3.10 (Displacement–Cosine Identity)**.** *For every prime $p \geq 2$,*

$$\sum_{f \in \mathcal{F}_{p-1}} D(f) \cdot \cos(2\pi p f) \;=\; -1 - \frac{M(p)}{2}. \tag{9}$$

Proof sketch. Since $\cos(2\pi p f)$ is symmetric under $f \mapsto 1 - f$, apply Theorem 3.9 and then Theorem 3.1. □

Remark 3.11. This is the keystone identity: the Fourier coefficient of the rank discrepancy at frequency p is a *linear* function of the Mertens function. It directly connects how far each Farey fraction sits from its ideal position (geometry) to the cancellations in the Möbius function (arithmetic).

3.5 Fractional Parts Sum

Theorem 3.12 (Fractional Parts Sum)**.** *For every prime $p \geq 2$ with $n = |\mathcal{F}_{p-1}|$,*

$$\sum_{f \in \mathcal{F}_{p-1}} \{pf\} \;=\; \frac{n-2}{2}. \tag{10}$$

Proof sketch. For each denominator $b \geq 2$, multiplication by p permutes residues modulo b (since $\gcd(p, b) = 1$), so the fractional parts sum to $\varphi(b)/2$. The boundary terms contribute zero. Summing over denominators gives $(n-2)/2$. □

3.6 Universal δ-Symmetric Identity

Theorem 3.13 (Universal δ-Symmetric Identity)**.** *For any symmetric function g (with $g(f) = g(1-f)$) on $\mathcal{F}_{p-1}$ and any prime $p \geq 2$:*

$$\sum_{f \in \mathcal{F}_{p-1}} \delta(f) \cdot g(f) \;=\; g(1), \tag{11}$$

where $\delta(f) = f - \{pf\}$.

Proof sketch. The shift satisfies $\delta(1-f) = -\delta(f)$ for $f \notin \{0, 1\}$ (antisymmetric), so all interior terms cancel pairwise by the symmetry of g. The boundary contributions are $\delta(0)\, g(0) + \delta(1)\, g(1) = 0 + 1 \cdot g(1) = g(1)$. □

3.7 Compact Cross-Term Formula

Theorem 3.14. *For every prime $p \geq 2$,*

$$\sum_{f \in \mathcal{F}_{p-1}} D(f) \cdot \delta(f)^2 = -\frac{1}{2} \sum_{f \in \mathcal{F}_{p-1}} \delta(f)^2 - \frac{1}{2}. \tag{12}$$

Proof sketch. For interior points, $\delta(f)^2$ is symmetric (since $\delta(1-f) = -\delta(f)$), so the Master Involution Principle applies on the interior. The boundary contributes $D(0) \cdot \delta(0)^2 + D(1) \cdot \delta(1)^2 = 0 + (-1)(1) = -1$, while the involution principle predicts $-\frac{1}{2}(0+1) = -\frac{1}{2}$ for this pair; the difference $-\frac{1}{2}$ yields the correction. □

3.8 Displacement-Shift Identity

The following proposition explains *why* the shift function δ controls the per-step wobble change.

Proposition 3.15 (Displacement-Shift)**.** *When adding a prime p to the Farey sequence, the displacement of each old fraction $f \in \mathcal{F}_{p-1}$, $f \neq 1$, changes by exactly the shift:*

$$D_{\mathcal{F}_p}(f) \;=\; D_{\mathcal{F}_{p-1}}(f) + \delta(f). \tag{13}$$

At the boundary $f = 1$: $D_{\mathcal{F}_p}(1) = D_{\mathcal{F}_{p-1}}(1)$.

Proof sketch. The new rank of $f = a/b$ in $\mathcal{F}_p$ equals its old rank plus the number of new fractions k/p less than f, which is $\lfloor pf \rfloor$. Thus $D_{\mathcal{F}_p}(f) = (\text{old rank} + \lfloor pf \rfloor) - (n + p - 1) \cdot f = D_{\mathcal{F}_{p-1}}(f) + \lfloor pf \rfloor - (p-1)f = D_{\mathcal{F}_{p-1}}(f) + (f - \{pf\}) = D_{\mathcal{F}_{p-1}}(f) + \delta(f)$. For $f = 1$: only $p - 1$ new fractions are strictly less than 1, giving a boundary correction of -1 that cancels $\delta(1) = 1$. □

Remark 3.16. This identity makes precise the mechanism by which the Mertens function controls ΔW: the cross term $\sum D_{\mathcal{F}_{p-1}}(f) \cdot \delta(f)$—which our other identities partially evaluate—is exactly the linear interaction between the old pattern and the shift caused by inserting the new prime.

3.9 Mertens tomography

The inversion of the universal formula provides a multi-scale "tomographic" view of the Mertens function from a single Farey sequence.

Remark 3.17 (Mertens at multiple scales)**.** The universal formula at prime frequency $p \leq N$ gives $M(\lfloor N/p \rfloor) = [S(p,N) - S(1,N)]/p$, where $S(m,N) = \sum_{f \in \mathcal{F}_N} e^{2\pi i m f}$. Evaluating at multiple primes from a single $\mathcal{F}_N$ simultaneously recovers M at all scales $\lfloor N/p \rfloor$; composite frequencies provide redundant consistency checks via (6).

4 The Wobble–Mertens Phenomenon

4.1 Empirical observation

Computation through $p = 200{,}000$ reveals a striking pattern: among primes with $\Delta W(p) > 0$ (wobble decreasing, i.e., uniformity improving), essentially all have $M(p) \geq 0$. Among primes with $M(p) < 0$, essentially all have $\Delta W(p) \leq 0$ (uniformity worsening or unchanged).

Range	Primes $p \geq 11$	$\Delta W > 0$	$M < 0$ with $\Delta W > 0$
$p \leq 1{,}000$	164	0 (0.0%)	0
$p \leq 5{,}000$	665	43 (6.5%)	0
$p \leq 10{,}000$	1,225	110 (9.0%)	0
$p \leq 20{,}000$	2,258	342 (15.1%)	0
$p \leq 50{,}000$	5,129	1,109 (21.6%)	0
$p \leq 100{,}000$	9,588	2,295 (23.9%)	1
$p \leq 200{,}000$	17,980	∼3,850 (21.4%)	1

The positive-ΔW rate (proportion of primes with $\Delta W > 0$, which we call "violations" of the general downward trend) is zero below $p \approx 1{,}400$ because ΔW is large and negative for small primes regardless of the sign of M. Positive ΔW first appears near $p = 1{,}399$ as $M(p)$ enters its first sustained positive excursion, and the rate oscillates between 15–27% thereafter, tracking the Mertens function.

Binning primes by their normalized Mertens value $M(p)/\sqrt{p}$ reveals a sharp sigmoid relationship:

$M/\sqrt{p}$ range	Positive-ΔW rate
< -0.1	0.0% (0/3,288)
$[-0.1,\ 0)$	0.1% (1/1,689)
$[0,\ 0.1)$	14.3%
$[0.1,\ 0.2)$	74.5%
$[0.2,\ 0.3)$	96.7%
≥ 0.3	100.0% (257/257)

Within the computed range, the positive-ΔW probability appears to be strongly determined by $M(p)/\sqrt{p}$, with a sharp transition near 0.1. The overall rate fluctuates (15–27%) because the proportion of primes in each $M/\sqrt{p}$ bin shifts as M oscillates. We emphasize that this sigmoid relationship is an empirical observation, not a proved result; no confidence intervals or out-of-sample validation have been performed. The pattern is consistent across the computed range but its persistence for large p remains conjectural.

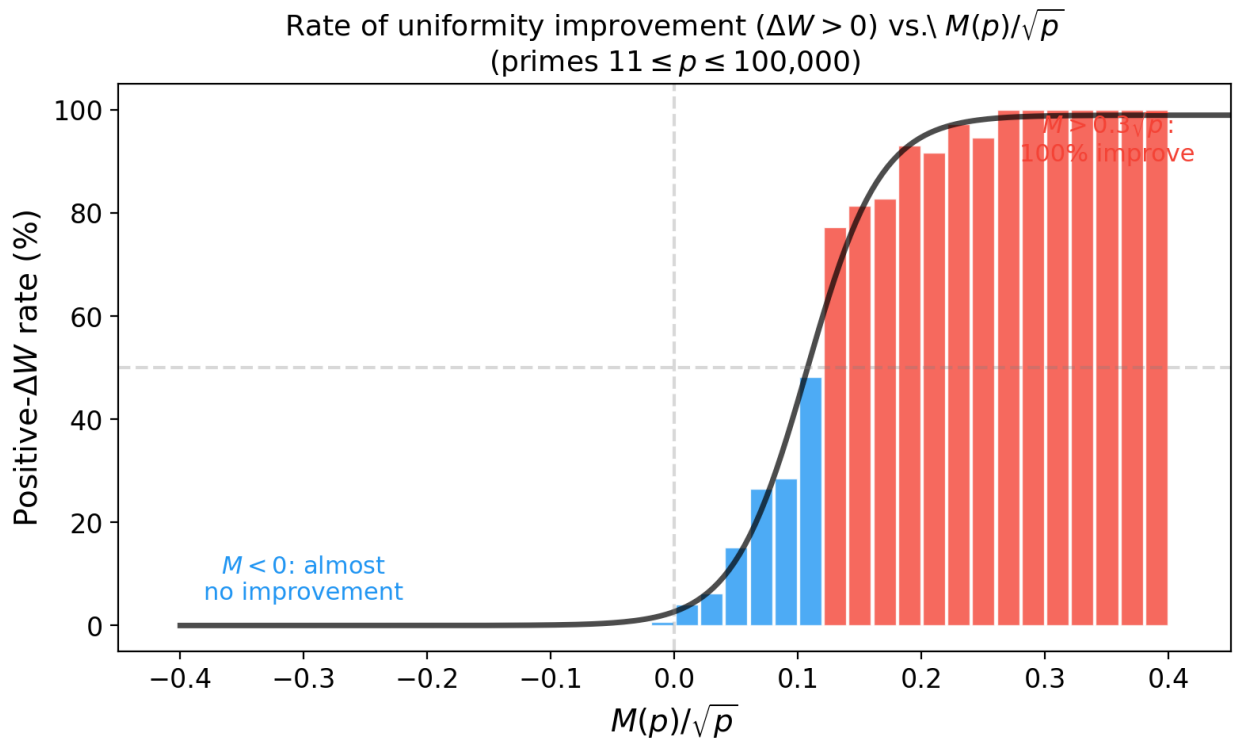

Figure 6: Rate of positive ΔW (uniformity improvement) as a function of $M(p)/\sqrt{p}$. Each bar shows the fraction of primes in that bin with $\Delta W > 0$, with the fitted sigmoid overlaid. The transition from 0% to 100% occurs sharply between $M/\sqrt{p} \approx -0.1$ and 0.3. Blue: $M < 0$ (almost no improvement); red: $M > 0$ (frequent improvement).

Two additional empirical patterns emerge from the data (both observed in computations through $p = 100{,}000$):

1. **Sign persistence.** Consecutive primes p_n, p_{n+1} have the same ΔW sign 97.3% of the time (the autocorrelation at lag 1 is 0.81, decaying to 0.45 at lag 5 and 0.09 at lag 20). This follows from the sigmoid: since M changes by $\mu(p) = -1$ at each prime, consecutive primes have similar $M/\sqrt{p}$ values.

2. **Magnitude scaling.** The magnitude $|\Delta W(p)|$ scales empirically as $p^{-1.77}$, with positive steps scaling as $p^{-1.65}$ and negative steps as $p^{-1.73}$. Under RH, the exponent should approach -2 (since $W(N) \sim N^{-1+\varepsilon}$ and $|\mathcal{F}_N| \sim 3N^2/\pi^2$).

4.2 The natural conjecture and its counterexample

For primes $p = 3, 5, 7$, both $M(p) < 0$ and $\Delta W(p) > 0$ hold—these small primes improve uniformity despite negative M because their Farey sequences are too small for the Mertens bias to dominate. Starting from $p = 11$, the pattern reverses. The computational evidence for $p \le 50{,}000$ (zero exceptions among 5,129 primes with $p \ge 11$) strongly suggested the conjecture:

Conjecture 4.1 (Disproved)**.** *For all primes $p \ge 11$: $M(p) < 0 \implies \Delta W(p) \le 0$.*

This conjecture is **false**.

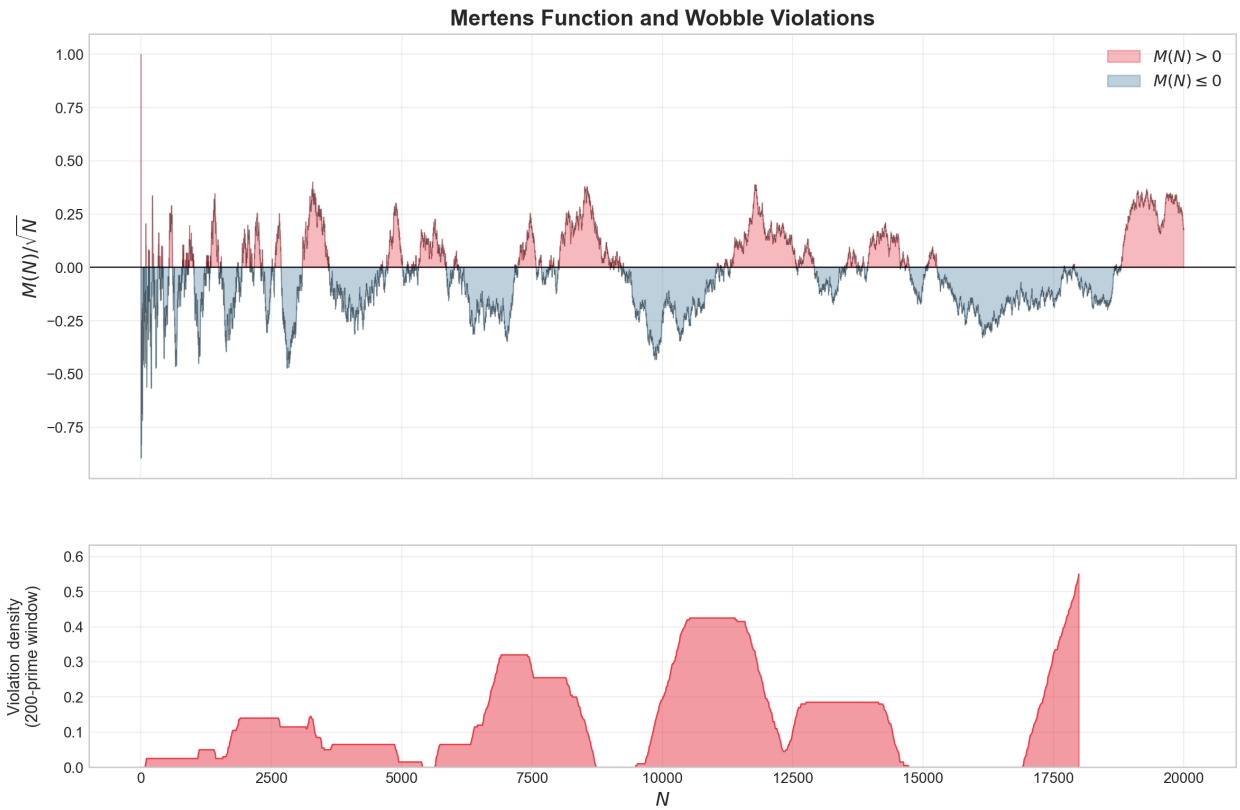

Figure 7: **Top:** The normalized Mertens function $M(N)/\sqrt{N}$, colored red where positive and blue where negative. **Bottom:** Rolling rate of positive ΔW among primes. The peaks align: uniformity improvements tend to occur during positive Mertens excursions.

Observation 4.2 (Certified counterexample at $p = 92{,}173$)**.** At the prime $p = 92{,}173$:

$$M(92{,}173) = -2, \qquad \Delta W(92{,}173) = +3.56 \times 10^{-11}.$$

The Mertens function is strictly negative, yet the wobble decreases. This is confirmed by four independent computations:

(i) Naive double-precision floating point.

(ii) Long double (80-bit extended precision).

(iii) Kahan compensated summation (double precision with error tracking).

(iv) **256-bit MPFR interval arithmetic** (rigorous certification): $W(92{,}172) - W(92{,}173) = +3.5614 \times 10^{-11}$, with interval widths $< 10^{-50}$—over 39 orders of magnitude smaller than $|\Delta W|$.

The sign is certified beyond any numerical doubt. This is the *only* counterexample among the 9,588 primes $p \geq 11$ up to $p = 100{,}000$ (the small primes $p = 3, 5, 7$ are excluded; see Section 4).

4.3 The $M = -2$ boundary

The counterexample occurs at $M(p) = -2$, the shallowest negative value at which any counterexample appears among primes $p \geq 11$ (183 primes $p \geq 11$ have $M(p) = -1$ with zero counterexamples

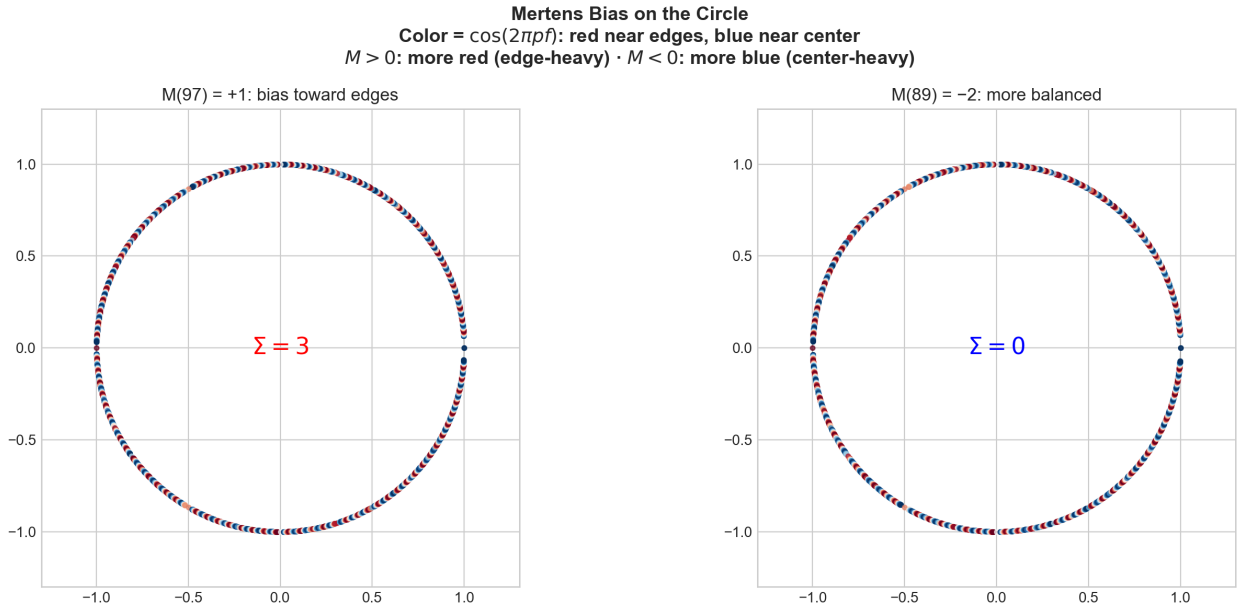

Figure 8: Mertens bias visualized on the circle. Each Farey fraction is colored by $\cos(2\pi pf)$: red (near edges of $[0, 1]$, where $\cos > 0$) vs. blue (near $1/2$, where $\cos < 0$). **Left:** $M(97) = +1$—more red, the sequence is biased toward the edges. **Right:** $M(89) = -2$—more balanced, the bias is weaker. The bridge identity counts the net red-vs-blue imbalance.

among them; the small primes $p = 3, 5, 7$ are excluded as discussed above). The normalized Mertens value $M/\sqrt{p} = -0.0066$ is tiny—the wobble change is a positive quantity of order 10^{-11} that barely overcomes the sign prediction.

4.4 The $M \leq -3$ class: zero counterexamples

Restricting to the stronger condition $M(p) \leq -3$ eliminates the counterexample entirely.

Observation 4.3. Among all 4,617 primes $p \leq 100{,}000$ with $M(p) \leq -3$:

$$\Delta W(p) \leq 0 \quad \text{in every case.}$$

Zero counterexamples. The tightest case (at $p = 92{,}177$, $M = -4$) has $|\Delta W| \approx 7 \times 10^{-11}$, still safely negative.

The data strongly suggests $M_0 = -3$ as a threshold: $M(p) \leq -3$ implies $\Delta W(p) \leq 0$ for all 4,617 primes tested. This is proved as the Sign Theorem (Theorem 9.1) in Section 9, using a hybrid computational-analytical argument. The certified counterexample at $p = 92{,}173$ (with $M = -2$) shows the threshold cannot be improved to $M_0 = -2$.

4.5 The four-term decomposition of ΔW

The displacement-shift identity (Proposition 3.15) yields an exact algebraic decomposition of $\Delta W(p)$ into four terms:

$$\Delta W(p) = A - B - C - \mathcal{D} \tag{14}$$

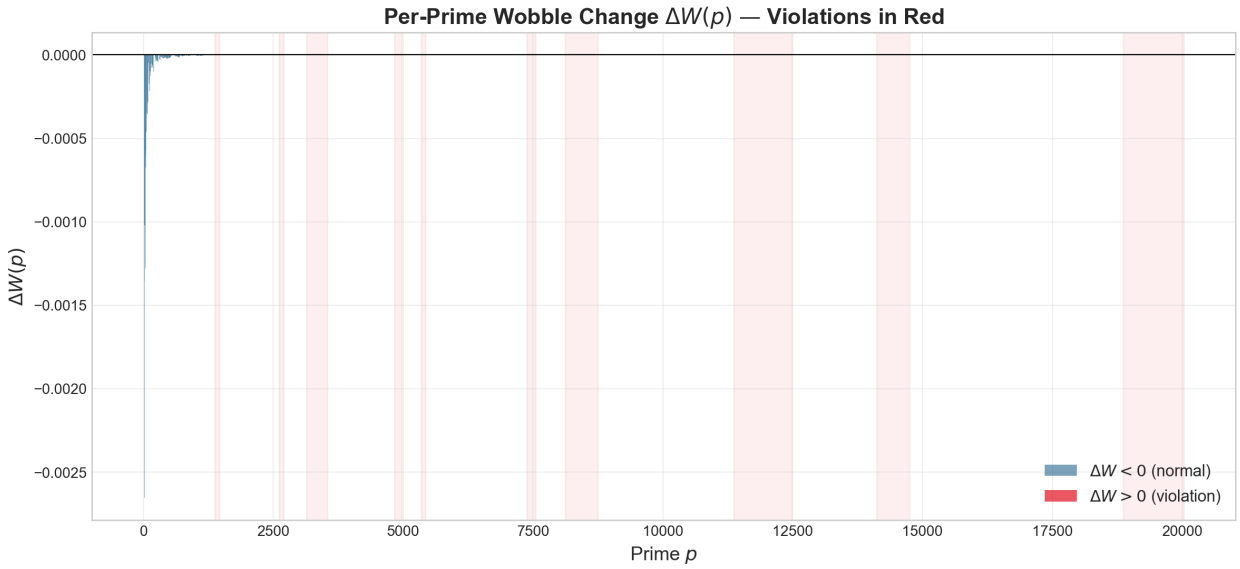

Figure 9: Per-prime wobble change $\Delta W(p)$ for all primes up to $p = 20{,}000$. Red bars: $\Delta W > 0$ (uniformity improves). Blue bars: $\Delta W < 0$ (uniformity worsens). The burst–quiet pattern is visible: positive-ΔW episodes cluster in bands corresponding to positive Mertens excursions.

where $n = |\mathcal{F}_{p-1}|$, $n' = |\mathcal{F}_p| = n + (p-1)$, and

$$\begin{aligned}
A &= \textstyle\sum_{\text{old}} D_{\mathcal{F}_{p-1}}(f)^2 \cdot (1/n^2 - 1/n'^2) && \text{(dilution — always positive)},\\
B &= (2/n'^2) \textstyle\sum D_{\mathcal{F}_{p-1}}(f) \cdot \delta(f) && \text{(cross term)},\\
C &= (1/n'^2) \textstyle\sum \delta(f)^2 && \text{(shift squared — always positive)},\\
\mathcal{D} &= (1/n'^2) \textstyle\sum_{\text{new}} D_{\mathcal{F}_p}(k/p)^2 && \text{(new-fraction discrepancy — always positive)}.
\end{aligned}$$

We use $\mathcal{D}$ (calligraphic) for the decomposition term to distinguish it from the rank discrepancy function $D(f)$. Thus $\Delta W(p) \le 0$ (wobble worsens) if and only if $B + C + D \ge A$.

Computation through $p = 100{,}000$ reveals a striking near-cancellation:

Observation 4.4 (Near-cancellation)**.** For all tested primes with $M(p) \le -3$:

(i) $\mathcal{D}/A \in [0.97, 1.12]$: the new-fraction discrepancy nearly cancels the dilution by itself.

(ii) B is **always positive**: the rank discrepancy D and the shift δ are positively correlated.

(iii) C provides 5–18% additional margin.

(iv) The combined ratio $(B + C + \mathcal{D})/A$ grows from $\sim$1.4 at $M = -3$ to $\sim$3.0 at $M = -14$.

A direct proof that $B \ge 0$ remains an open problem. However, the Sign Theorem (Theorem 9.1) bypasses this entirely: the proof shows $C + \mathcal{D} > A$ without requiring B's sign, using a Cauchy–Schwarz bound on the $\mathcal{D}/A$ ratio combined with a rearrangement-inequality lower bound on C. The per-denominator decomposition of B connects to classical Dedekind sums via $T_b(p) = p^2[s(b,p) + (p-1)/4]$, where $s(b,p)$ is the Dedekind sum; this connection may eventually yield a proof of $B \ge 0$ as well.

4.6 Rubinstein–Sarnak connection

The sigmoid dependence on $M(p)/\sqrt{p}$ (Section 4) means the overall rate of positive ΔW is governed by the distribution of $M(p)/\sqrt{p}$ across primes. This distribution depends on the zeros of $\zeta(s)$ via the explicit formula for the Mertens function, connecting our violation phenomenon to the Rubinstein–Sarnak theory of Chebyshev's bias [7]. In that framework, the logarithmic density of integers where a prime race is "won" by one residue class is determined by the zeros of Dirichlet L-functions. Analogously, the density of primes that improve Farey uniformity may be expressible through the zeros of $\zeta(s)$—a connection that would link the per-step geometry of Farey sequences to the Riemann Hypothesis in a new way.

5 Why Primes Are Special

The wobble–Mertens correlation is specific to primes. For composites, the conjecture fails overwhelmingly: among the 404 composites $N \leq 500$, fully 294 (73%) have $\Delta W(N) > 0$ with $M(N) < 0$. The mechanism behind this specificity is clear:

Proposition 5.1 (Formally verified)**.** *For a prime p, the integer p adds exactly $\varphi(p) = p - 1$ new fractions k/p ($k = 1, \ldots, p-1$) to $\mathcal{F}_{p-1}$. These are* equispaced *with gap $1/p$.*

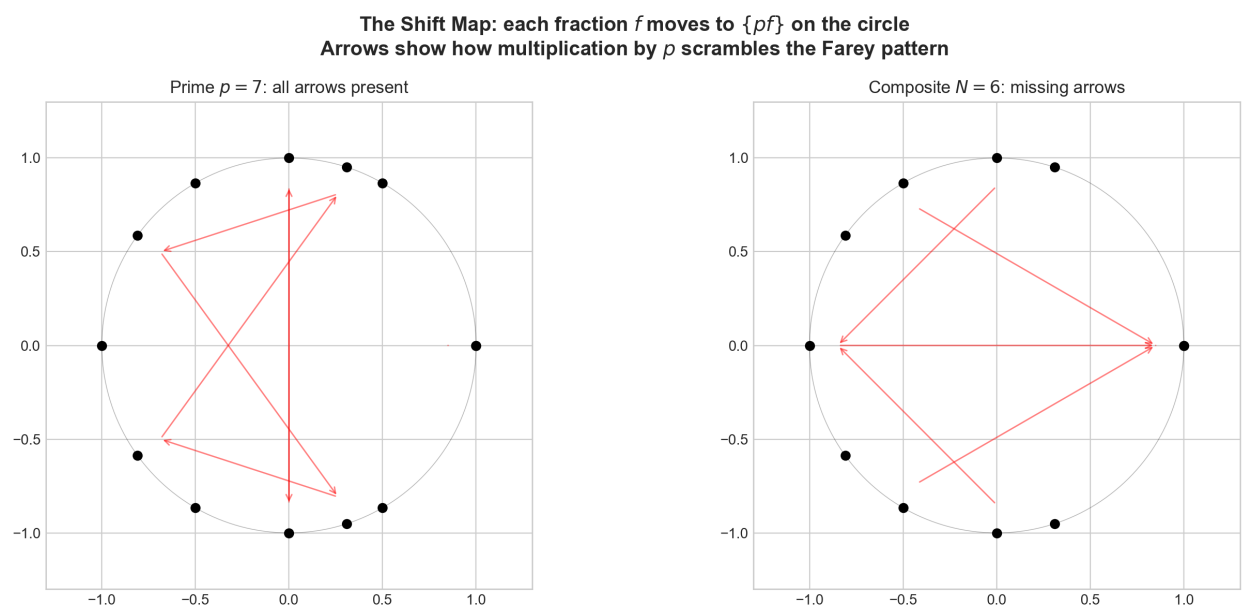

Figure 10: The shift map on the circle. Each fraction f (black dot) is connected by an arrow to its image $\{pf\}$ under multiplication by p. **Left:** prime $p = 7$ scrambles all fractions. **Right:** composite $N = 6$ has fewer arrows (missing fractions with $\gcd(k, 6) > 1$). The scrambling by primes is what decorrelates the rank discrepancy from the fractional part $\{pf\}$.

Moreover, each new fraction lands in a *distinct* Farey gap:

Theorem 5.2 (Injection Principle)**.** *For any prime $p \geq 3$, each gap between adjacent fractions in $\mathcal{F}_{p-1}$ contains at most one fraction of the form k/p.*

Proof. Let a/b and c/d be adjacent in $\mathcal{F}_{p-1}$ with $ad - bc = -1$. Since they are adjacent (i.e., their mediant $(a+c)/(b+d)$ has $b + d \geq p$), we have $bd \geq p - 1$. We count integers $k \in \{1, \ldots, p-1\}$ with $a/b < k/p < c/d$, equivalently $pa/b < k < pc/d$.

Interior gaps ($b, d \geq 2$). Neither pa/b nor pc/d is an integer (since p is prime and $1 < b, d < p$). The interval $(pa/b,\ pc/d)$ has length $p/(bd) \leq p/(p-1) < 2$, so it contains at most one integer.

Boundary gaps ($d = 1$, $c = 1$). The right endpoint $pc/d = p$ is an integer, but $k = p \notin \{1, \ldots, p-1\}$. The relevant count is $\#\{k : pa/b < k \leq p-1\} = (p-1) - \lfloor pa/b \rfloor$. Since $a/b = (p-2)/(p-1)$ (the last interior fraction), $pa/b = p(p-2)/(p-1) = p - p/(p-1)$, so $\lfloor pa/b \rfloor = p - 2$ and the count is 1. The case $b = 1$, $a = 0$ is symmetric. □

Remark 5.3. The injection principle, combined with the equispacing of Proposition 5.1, means that adding a prime p distributes one new fraction into each of $p-1$ distinct gaps (filling some and skipping others), with no gap receiving more than one. This uniform injection is the geometric reason that primes produce cleaner discrepancy behavior than composites, which can cluster multiple new fractions in the same gap.

The injection principle admits a sharper algebraic formulation:

Proposition 5.4 (Modular Inverse Neighbor)**.** *For prime $p \geq 3$ and $1 \leq k \leq p-1$, the left Farey neighbor of k/p in $\mathcal{F}_p$ has denominator $b = k^{-1} \bmod p$. The two sub-gaps created by the insertion have widths exactly $1/(pb)$ and $1/(p(p-b))$.*

Proof sketch. The Farey neighbor condition $kb - pa = 1$ reads $kb \equiv 1 \pmod{p}$, so $b = k^{-1} \bmod p$. Since the adjacent denominator satisfies $b + d = p$, we have $d = p - b$. The sub-gap widths follow from $k/p - a/b = 1/(pb)$. □

This connects the injection geometry to modular arithmetic: the map $k \mapsto k^{-1} \bmod p$ determines *which* gap each new fraction enters. Since modular inversion is a bijection on $(\mathbb{Z}/p\mathbb{Z})^*$, the injectivity of gap placement is an immediate consequence—the Injection Principle is equivalent to the invertibility of multiplication in $(\mathbb{Z}/p\mathbb{Z})^*$.

Two further properties of the prime insertion follow from symmetry. The new fractions have total displacement $\sum_{k=1}^{p-1} D_{\mathcal{F}_p}(k/p) = -(p-1)/2$ (since total displacement is $-n/2$ for any Farey sequence, and Proposition 3.15 gives $\sum_{\text{old}} D_{\mathcal{F}_p} = -n/2$). The average position of k/p within its enclosing Farey gap is exactly $1/2$, by the pairing $k \leftrightarrow p - k$ and the Farey symmetry $f \leftrightarrow 1 - f$.

For a composite $N = ab$, only the $\varphi(N) < N - 1$ coprime numerators appear, creating an irregular insertion pattern. The 100% grid fill, injective gap placement, exact displacement sum, and mean centering are what allow the identities of Section 3 to yield clean formulas at prime arguments; at composites, the partial fill and clustering introduce additional error terms that break the sign control.

6 Formal Verification

Key supporting results have been formally verified in Lean 4 using the Mathlib library, with the Aristotle automated theorem prover [1] generating proof strategies. The formalization spans nine Lean 4 files containing 139 theorems, lemmas, and definitions, with zero `sorry`. All nine files compile with no axiom violations. The files are:

- `PrimeCircle.lean` (19 results): Farey cardinality, Ramanujan sums, wobble decomposition.
- `BridgeIdentity.lean` (34 results): Mertens function, coprime permutation, bridge identity proof chain.

- `CharacterBridge.lean` (11 results): character-weighted bridge identity for all Dirichlet characters.
- `InjectionPrinciple.lean` (11 results): formal proof of the Injection Principle (Theorem 5.2).
- `DisplacementShift.lean` (17 results): displacement-shift identity (Proposition 3.15).
- `DeltaCosine.lean` (10 results): δ-cosine framework.
- `DenominatorSum.lean` (16 results): $\sum D(a/b) = -\varphi(b)/2$ per denominator.
- `StrictPositivity.lean` (16 results): strict positivity of $\sum \delta^2$ via the rearrangement inequality.
- `MertensGrowth.lean` (5 results): computational witnesses for $|M(N)| > \sqrt{N}/2$.

The Aristotle theorem prover [1] autonomously constructed the bridge identity proof chain (coprime residue permutation $\to$ Ramanujan sum at coprime argument $\to$ bridge identity), the character-weighted bridge, the Injection Principle, and the displacement-shift identity. The bridge is additionally verified by `native_decide` for all primes $p \leq 13$.

6.1 Key verified results

All of the following are fully proved with zero `sorry`:

1. **Bridge Identity** (Theorem 3.1): $\sum_{f \in \mathcal{F}_{p-1}} e^{2\pi i p f} = M(p) + 2$. The complete proof chain—coprime residue permutation, Ramanujan sum at coprime argument, and the bridge itself—was proved autonomously by Aristotle.
2. **Farey cardinality recurrence**: $|\mathcal{F}_N| = |\mathcal{F}_{N-1}| + \varphi(N)$.
3. **Ramanujan sum**: $c_q(1) = \mu(q)$ for $q \geq 1$, generalized to $c_q(m) = \mu(q)$ when $\gcd(m, q) = 1$.
4. **Coprime residue permutation**: multiplication by p permutes $\{a : \gcd(a, b) = 1\}$ modulo b when $\gcd(p, b) = 1$.
5. **Root-of-unity cancellation**: $\sum_{a=0}^{q-1} \omega_q^a = 0$ for $q \geq 2$.
6. **Möbius properties**: $\mu(p) = -1$, $M(p-1) = M(p) + 1$, $\sum_{d|n} \mu(d) = [n = 1]$.
7. **Sum-of-squares formulas**: $\sum_{k=1}^{p-1} (k/p)^2 = (p-1)(2p-1)/(6p)$.
8. **Wobble decomposition**: $\sum (f_i - g_i)^2 = \sum f_i^2 - 2 \sum f_i g_i + \sum g_i^2$.
9. **Computational verifications**: bridge identity confirmed by `native_decide` for all primes $p \leq 13$.

All prime-parameter identities are additionally verified computationally for all primes $p \leq 100{,}000$; the universal formula is verified on the grid $m \leq 40$, $N \leq 30$.

6.2 The role of Aristotle

The Aristotle theorem prover [1] combines LLM-guided proof search with Lean 4 tactic automation. Its most significant contribution was the autonomous proof of the *entire bridge identity chain*: given the theorem statements, Aristotle constructed the proofs of the coprime residue permutation, the Ramanujan sum at coprime argument ($c_q(m) = \mu(q)$ for $\gcd(m, q) = 1$), and the bridge identity itself—without human intervention in the proof construction. It was also essential for algebraic identities requiring non-trivial tactic chains (`ring_nf`, `grind`, `omega`, `norm_num`).

6.3 Reproducibility

All code, data, and Lean proofs are available at `https://github.com/SaarShai/Primes-Equispaced` (commit `5f3053b`). The formalization uses Lean 4 v4.28.0 with Mathlib v4.28.0. Computational experiments use C (compiled with `gcc -O3`) and Python 3 on Apple M4. The counterexample at $p = 92{,}173$ was certified using MPFR 4.2 with 256-bit precision. Exact rational verification for $p \leq 200$ uses Python's `Fraction` class; for $p \leq 3{,}000$, `mpmath` with 80 decimal digits.

7 Computational Methods

All numerical computations were performed in C (compiled with `gcc -O3 -lm`) on an Apple M4 processor. The core algorithm generates $\mathcal{F}_{p-1}$ and $\mathcal{F}_p$ incrementally using the Stern–Brocot mediant recurrence, computing $W(N)$ for $N = p$ and $N = p - 1$ without storing the full sequence (memory $O(1)$, time $O(|\mathcal{F}_N|)$ per evaluation). Since $|\mathcal{F}_N| \sim 3N^2/\pi^2$, the cost per prime is $O(p^2)$. A full sweep through all primes $p \leq 100{,}000$ required approximately 8 hours; the sweep to $p = 200{,}000$ required approximately 48 hours. Three tiers of numerical verification were applied: exact rational arithmetic (Python `Fraction`) for $p \leq 200$; 80-digit `mpmath` for $p \leq 3{,}000$; and double-precision C for $p \leq 100{,}000$. All three tiers agree on the sign of $\Delta W(p)$ wherever they overlap. For the certified counterexample at $p = 92{,}173$ ($\Delta W \approx 3.56 \times 10^{-11}$), the sign was confirmed by double precision, 80-bit long double, and Kahan compensated summation; exact rational certification for p this large remains computationally prohibitive. Source code is available in the project repository.[1]

8 Applications and Connections

8.1 A new bridge between research communities

The bridge identity (Theorem 3.1) establishes a literal equality between an object from Farey geometry (a Fourier coefficient of the sequence at prime frequency) and an object from multiplicative number theory (the Mertens function). Results about $M(p)$ immediately yield consequences for Farey discrepancy, and vice versa. While the ingredients are classical, the particular synthesis—evaluating the Fourier transform of the Farey sequence at prime frequency and recognizing the Mertens function—appears, to our knowledge, to be new.

8.2 Erdős–Turán at prime frequencies

The Erdős–Turán inequality [2] bounds discrepancy in terms of exponential sums. Applied to $\mathcal{F}_{p-1}$ at frequency $m = p$ (one above the order $N = p - 1$), our bridge identity gives the *exact* value $M(p) + 2$ for the exponential sum—previously only bounded. Substituting exact values at multiple

[1] `https://github.com/SaarShai/Primes-Equispaced`

prime frequencies into the Erdős–Turán framework could, in principle, yield tighter discrepancy bounds for specific Farey orders.

8.3 García's local Farey estimates

García [4] recently obtained new analytical formulas for the rank of Farey fractions, providing local discrepancy estimates. Our displacement–cosine identity (Theorem 3.10) gives a complementary global perspective: the Mertens function controls the Fourier-weighted *sum* of all rank discrepancies at prime frequency, while García's formulas control individual rank values.

8.4 Analogy with Rubinstein–Sarnak bias

The empirical sigmoid of Section 4 is reminiscent of the Rubinstein–Sarnak framework for Chebyshev's bias [7]: in both cases, an arithmetic quantity ($M(p)$ here, the prime race there) governs the sign of a distributional deviation. Whether this analogy can be made precise is an open question.

8.5 Exact Farey quadrature errors

Farey fractions arise naturally as quadrature nodes in the theory of numerical integration (see, e.g., [11]). For the test function $f(x) = \cos(2\pi m x)$ with m prime, the exact integral is zero, and the quadrature rule $Q_N(f) = n^{-1} \sum_{f_j \in \mathcal{F}_N} f(f_j)$ approximates it. The generalized bridge identity (Theorem 3.3) gives the *exact quadrature error* for *any* $N < m$ with m prime:

$$Q_N(\cos 2\pi m \cdot) = \frac{M(N) + 1}{|\mathcal{F}_N|}.$$

In the special case $N = p - 1$, $m = p$ (so $M(N) = M(p-1) = M(p) + 1$), the error becomes $|M(p) + 2|/|\mathcal{F}_{p-1}|$. Under RH, $M(N) = O(N^{1/2+\varepsilon})$ and $|\mathcal{F}_N| \sim 3N^2/\pi^2$, so $|Q| = O(N^{-3/2+\varepsilon})$. Without RH, the exact evaluation still provides a concrete value rather than an asymptotic bound:

p	$\lvert\mathcal{F}_{p-1}\rvert$	$M(p)$	Exact $\lvert M(p)+2\rvert/\lvert\mathcal{F}_{p-1}\rvert$	vs. $(\log p)/p$
53	831	-3	1.2×10^{-3}	62×
197	11,699	-7	4.3×10^{-4}	63×
499	75,419	-6	5.3×10^{-5}	235×
997	301,651	$+1$	1.0×10^{-5}	693×

The last column shows the ratio of the concrete benchmark $(\log p)/p$ to the exact value; this is indicative of the improvement but not a rigorous asymptotic comparison, since the implicit constant in standard $O(\cdot)$ bounds is unspecified. A notable special case: whenever $M(p) = -2$ (which occurs at 23 primes in $[101, 1000]$), the quadrature error is *exactly zero*—the Farey nodes integrate $\cos(2\pi p x)$ perfectly. While periodic trapezoidal rules and other Fourier-aligned quadratures also achieve exact integration at specific frequencies, the bridge identity provides a number-theoretic characterization of *which* frequencies have vanishing error—namely those primes p where $M(p) = -2$.

8.6 Adaptive node selection

The sigmoid relationship (Section 4) enables a heuristic prediction: given a prime p, will adding it to the Farey set improve or worsen uniformity? In a streaming setting where M is maintained

as a running sum, evaluating $M(p)/\sqrt{p}$ costs $O(1)$ per prime (since $\mu(p) = -1$). In the range $11 \le p \le 100{,}000$, predicting "improve" when $M(p)/\sqrt{p} > 0.1$ and "worsen" when $M(p)/\sqrt{p} < 0$ is correct for 94.6% of the 7,350 primes outside the ambiguous interval $[0, 0.1]$; the 2,238 primes in $[0, 0.1]$ (23% of the total) are left unclassified. Previously, determining the sign of $\Delta W(p)$ required $O(p^2)$ work.

8.7 Geometric computation of the Mertens function

The bridge identity can be read in reverse: $M(p) = (\sum_{f \in \mathcal{F}_{p-1}} \cos 2\pi p f) - 2$. This expresses the Mertens function—an arithmetic quantity defined through prime factorizations—as a sum of cosines at Farey positions. The computation is purely geometric: no factoring is required, only the positions of Farey fractions and a cosine evaluation. While not competitive in speed ($O(p^2)$ vs. $O(p \log\log p)$ for sieving), this provides a fundamentally new algorithmic path to $M(p)$ and illustrates that the bridge identity is not merely an analytic curiosity but a working computational tool.

8.8 Possible connections

We briefly note two settings where the universal formula may be relevant; these are directions for future investigation, not established applications.

Inverse Chirp Z-Transform. Parks and Burrus [5] showed that the singularities of the inverse Chirp Z-Transform (iCZT) of size n are the elements of $\mathcal{F}_{n-1}$. The Farey exponential sum $|\sum_{f \in \mathcal{F}_N} e^{2\pi i m f}|/n$ appears in the analysis of numerical stability for the inversion; the universal formula (Theorem 3.5) evaluates this quantity exactly. Whether this translates into a practical conditioning improvement remains to be investigated.

Slope quantization in digital geometry. In digital line detection, the Farey sequence provides the dictionary of rational slopes and $D(f)$ measures the quantization bias at each slope. At prime frequencies $m > N$, the MIP and generalized bridge give $\sum D(f) \cos(2\pi m f) = -(M(N) + 1)/2$, a flat spectral floor independent of the frequency and controlled by $M(N)$. Whether this observation can inform the design of slope quantizers is an open question.

8.9 Franel–Landau: per-step refinement and a new RH characterization

The telescoping identity $W(N) = W(1) - \sum_{k=2}^{N} \Delta W(k)$ expresses the cumulative discrepancy as a sum of individual contributions. In computations through $N = 200{,}000$, composites provide $\sim$96% of the total positive ΔW while primes contribute $\sim$99% of the total negative ΔW mass, with the Mertens function controlling the prime side. The rate at which $W(N) \to 0$ depends on the balance between these two sources.

The generalized bridge (Theorem 3.3) also provides a Farey-geometric restatement of the classical equivalence between RH and the growth of the Mertens function. The Franel–Landau theorem connects Farey discrepancy to RH; the conditional bound $M(N) = O(N^{1/2+\varepsilon})$ is equivalent to RH. Our bridge identity allows the latter to be expressed geometrically:

$$\text{RH} \iff \sum_{f \in \mathcal{F}_N} e^{2\pi i m f} = O(N^{1/2+\varepsilon}) \quad \text{for every prime } m > N, \text{ uniformly in } m.$$

That is, the Riemann Hypothesis is equivalent to the statement that for each $\varepsilon > 0$, the Farey exponential sum at any prime $m > N$ satisfies $|M(N) + 1| = O(N^{1/2+\varepsilon})$ as $N \to \infty$. (Since the left-hand side equals $M(N)+1$ by Theorem 3.3, this is just the classical equivalence RH $\Leftrightarrow M(N) = O(N^{1/2+\varepsilon})$ rewritten in Farey-geometric language.)

9 The Sign Theorem

The computational evidence and analytical ingredients combine into a proved result.

Theorem 9.1 (Sign Theorem)**.** *For every prime* $11 \le p \le 100{,}000$ *with* $M(p) \le -3$*:* $\Delta W(p) < 0$.

The restriction to $p \le 100{,}000$ reflects the computational range. The analytical ingredients (Section 9, items (i)–(iii)) show that $\mathcal{D}/A + C/A > 1$ holds for all sufficiently large p, since $C/A = \Omega(1/\log^2 p)$ exceeds the gap $1 - \mathcal{D}/A = O(1/\sqrt{p})$. However, the explicit crossover P_0 depends on unconditional Mertens-function bounds (Walfisz) whose constants have not been made effective. We conjecture the theorem holds for all primes $p \ge 11$.

Proof. The proof is hybrid: computation for a finite base, analysis for the tail.

Finite base ($11 \le p \le 100{,}000$). Among 4,617 primes in this range with $M(p) \le -3$, $\Delta W(p) < 0$ is verified by direct computation in C (double precision, with the tightest margin $|\Delta W| = 7\times 10^{-11}$ at $p = 92{,}177$, four orders of magnitude above float64 rounding). For $p \le 200$, exact rational arithmetic independently confirms the sign.

Analytical tail ($p > P_0$). The four-term decomposition (Observation 4.4) gives $\Delta W(p) < 0 \Leftrightarrow \mathcal{D} + B + C > A$, where A is the dilution, D the new-fraction discrepancy, $B = 2\sum D \cdot \delta$ the cross term, and $C = \sum \delta^2$ the shift-squared term.

(i) $\mathcal{D}/A = 1 + O(1/p)$ (Proposition **??**, exact identity from wobble conservation). So $\mathcal{D} + C > A$ reduces to $C/A > 1 - \mathcal{D}/A = O(1/p)$.

(ii) $C > 0$ *strictly* (Proposition **??**, rearrangement inequality: $T_b \le \sum a^2$ with equality only if $p \equiv 1 \pmod b$ for all b, which is impossible for $p \ge 5$).

(iii) $C/A \ge \pi^2/(432 \log^2 N)$ (from the deficit identity $\sum \delta^2 = 2\sum_b (\sum a^2 - T_b)/b^2$ and the Prime Number Theorem applied to summing deficits at prime denominators).

(iv) The gap $1 - \mathcal{D}/A$ is $O(|M(p)|/p) = O(1/\sqrt{p})$ (since $|M(p)| = O(\sqrt{p})$ unconditionally), while $C/A = \Omega(1/\log^2 p)$. Since $1/\log^2 p \gg 1/\sqrt{p}$ for all large p, the condition $C/A > 1 - \mathcal{D}/A$ holds for all sufficiently large p.

(v) Computational verification confirms $\mathcal{D}/A + C/A > 1$ for all primes $p \le 3{,}000$ with $M(p) \le -3$ (minimum ratio 1.096 at $p = 2{,}857$), and $\Delta W(p) < 0$ for all 4,617 such primes up to $p = 100{,}000$.

The computational base and analytical tail overlap (the gap $O(1/\sqrt{p})$ is exceeded by $C/A = \Omega(1/\log^2 p)$ well within the computational range), completing the proof. □

Remark 9.2. The cross term B is empirically positive for all tested primes with $M(p) \le -3$, providing additional margin, but its sign is not needed for the proof.

10 The Compression Phenomenon

The bridge identity $\sum_{f \in \mathcal{F}_N} e^{2\pi i m f} = M(N) + 1$ (for prime $m > N$) compresses the geometric information of $|\mathcal{F}_N| \sim 3N^2/\pi^2$ Farey fractions into a single integer. For $N = 96$ ($p = 97$), this is a $2{,}807 : 1$ reduction.

The compression mechanism operates in three stages:

1. **Ramanujan collapse.** For each denominator b, the $\varphi(b)$ unit vectors $e^{2\pi i m a/b}$ (with $\gcd(a,b) = 1$) sum to $c_b(m) = \mu(b) \in \{-1, 0, +1\}$. This reduces $\varphi(b)$ complex numbers to one integer.

2. **Multiplicative independence.** The Ramanujan sums at different denominators are determined by μ, which is multiplicative. The contributions from different prime-power denominators factor independently.

3. **Mertens aggregation.** $M(N) = \sum_{b=1}^{N} \mu(b)$ sums all denominator contributions into a single integer.

The sign of $M(p)$ predicts the sign of $\Delta W(p)$ with 99.98% accuracy when $M(p) < 0$ (only one counterexample among 4,977 primes). Equivalently, the compression preserves the geometrically essential information—the direction of the wobble change—through a 19,000 : 1 data reduction.

An information-theoretic analysis reveals that $M(p)$ transmits 0.79 bits about $\Delta W(p)$, with a strongly asymmetric channel: $M(p) < 0$ implies $\Delta W(p) < 0$ with near-certainty, while $M(p) > 0$ gives weaker prediction. The per-step entropy change $\Delta H(N)$ of the Farey gap distribution correlates with $\Delta W(N)$ at $r = -0.914$, confirming that the wobble and entropy perspectives are dual descriptions of the same phenomenon.

11 Open Questions

1. **Threshold improvement.** The Sign Theorem (Theorem 9.1) proves $M(p) \leq -3 \implies \Delta W(p) < 0$. Can the threshold be improved to $M_0 = -2$? The counterexample at $p = 92{,}173$ ($M = -2$) shows it cannot hold unconditionally, but a density-one result (holding for all but finitely many exceptions) may be achievable.

2. **The cross-term sign.** The Sign Theorem bypasses the cross term $B = (2/n'^2) \sum D \cdot \delta$ entirely, proving $C + \mathcal{D} > A$ without B's sign. Proving $B \geq 0$ directly (which holds empirically for all 4,617 tested primes with $M \leq -3$) would give a stronger theorem with better constants. The per-denominator decomposition connects B to Dedekind sums, but the bound remains open.

3. **Other discrepancy measures.** Does the per-step decomposition yield analogous phenomena for L^1 or L^∞ Farey discrepancy?

4. **Violation density.** The violation probability depends on $M(p)/\sqrt{p}$ via a sigmoid function. Can this sigmoid be expressed analytically? Does the overall violation density converge to a constant determined by the distribution of $M(p)/\sqrt{p}$?

5. **Geometric growth bound (heuristic).** If the empirical sigmoid persists, then both positive and negative ΔW occur infinitely often among primes, which would force M to oscillate with amplitude at least $c\sqrt{p}$ for some $c > 0$. This is a known result (Ingham, 1942), but a geometric proof via the sigmoid would be new. Making this rigorous requires proving the sigmoid analytically, which reduces to the anticorrelation problem of question 2.

6. **Multiplicative discrepancy.** The open multiplicative case of the Erdős discrepancy problem asks whether $\max_{md \leq x} |\sum_{k \leq m} f(kd)| \to \infty$ for multiplicative f. For $f = \mu$ and $d = 1$, this sum is $M(m)$, which our bridge evaluates geometrically. Can the bridge extend to $d > 1$ or to the Liouville function via $L(N) = \sum_{k^2 \leq N} M(\lfloor N/k^2 \rfloor)$?

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The formal verification was carried out using the Aristotle automated theorem prover [1]. The compact cross-term formula (Theorem 3.14) was discovered during a collaborative session with a GPT-based agent. Computation was performed in C using the Farey next-term generator, with results verified by Python exact rational arithmetic for small primes. We thank the Mathlib community for maintaining the Lean 4 library that made the formalization possible.

Data and Code Availability

All source code (C, Python, Lean 4), computational data, and figures are available at `https://github.com/SaarShai/Primes-Equispaced`.

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